⭐ STEP 1 — MVP FEATURE LIST (REALISTIC, BUILDABLE, HIGH-IMPACT)

This MVP is intentionally *lean*, but powerful enough to blow every competitor away.

I'm giving you a **Tier 1 (must-have)** and **Tier 2 (lightweight enhancements)** structure so you don't overbuild.

✅ Tier 1 — MVP MUST-HAVES

1. Onboarding → Choose Your Focus

- LDL Mode
- Triglyceride Mode
- "Both" option

This instantly personalizes your app more than any competitor.

2. Lab Entry (Manual + Photo Import MVP-ready)

- Enter LDL, HDL, Triglycerides, Non-HDL, Total
- Date picker
- Tag as fasting / non-fasting
- **OCR-ready placeholder (the UI exists; OCR can be activated later)**

You can start without OCR but lay the groundwork.

3. Cholesterol Score (Core Value Prop)

This is your "Credit Score of Heart Health."

A single 0–100 score derived from:

- LDL
- HDL
- TG
- Non-HDL
- Total/HDL ratio
- Behavior score (lightweight)

This is the emotional hook.

4. Daily Behavior Tracker (Tap-Based, Fast)

Daily taps:

- Alcohol (0 / 1 / 2+ drinks)
- Steps goal hit (yes/no)
- Late-night eating (yes/no)
- Heavy carbs (yes/no)
- High saturated fat day (yes/no)
- Stress (low/med/high)
- Sleep (<6h, 6–8h, >8h)
- Medication taken (yes/no)

These are clinically meaningful for LDL/TG.

Each tap updates a hidden "Daily Heart Score" that informs insights.

5. Trends Dashboard

- Cholesterol Score over time
- LDL/TG charts
- Average behavior score
- Simple insights ("Weeks with fewer than 2 alcohol days → better TG")

Visually simple.
No clutter.

6. Weekly Summary (AI Text or Templated Text)

A clean card showing:

- Best habits
- Worst habits
- Lab trends
- Predicted trajectory

"Here's what improved your TG this week."
This is where the *magic* happens.

7. Minimal Food Guidance

Just a basic lookup:

- Good choices for LDL
- Good choices for TG
- Avoid list
- Better swaps ("Instead of ___ try ___")

No huge database.
Keep MVP tight.

8. Settings

- Reminders: "Log habits," "Log meds," "Upcoming labs"
- Export data
- Switch LDL/TG mode

That's your MVP.
Enough to launch. Enough to solve a real problem. Not bloated.

🟦 Tier 2 — Lightweight Enhancements (Add Weeks 3–6 After MVP)

- OCR full lab extraction
- Apple Health integration (pull steps, weight, heart rate)
- Medication list + adherence tracker
- Goal planning ("Lower LDL by 20 points")
- Risk Score (ASCVD simplified)
- Recipe recommendations
- Alcohol tracker calendar
- Weight trend correlation

But these are AFTER launch.

⭐ STEP 2 — CHOLESTEROL SCORE (HOW WE CALCULATE IT)

This is your *intellectual property*.
Simple, clear, not medical advice — but *directionally meaningful*.

The score is composed of:

- **50% Lab Metrics**
- **30% Behavior Score**
- **20% Trend Direction (improving vs worsening)**

Let's break it down:

🧪 A. Lab Score (0–50 points)

We normalize each metric to a 0–10 subscore:

- LDL: 0–10
- HDL: 0–10
- TG: 0–10
- Non-HDL: 0–10
- Ratio (Total/HDL): 0–10

Then we weight them by clinical importance:

- LDL: 35%
- TG: 25%
- HDL: 15%

- Ratio: 15%
- Non-HDL: 10%

This produces a lab score out of 50.

🍏 B. Behavior Score (0–30 points)

Based on:

- Steps
- Alcohol
- Sleep
- Late eating
- High-carb day
- Saturated fat
- Medication adherence
- Stress

We assign +1 or –1 daily, then compute weekly averages.

This rewards consistency without nagging.

📈 C. Trend Score (0–20 points)

Measures whether LDL/TG are moving:

- ↓ improving = +10 to +20
- stable = +8 to +10
- ↑ worsening = 0 to +5

This tells the user "the direction is good/bad."

Final Score = Lab (50) + Habits (30) + Trend (20)

A perfect **100** means you're killing it.
A score of **60–70** means "room for improvement."

Massive USP.

⭐ STEP 3 — HABIT → LIPID IMPACT MODEL

This is the secret engine that drives:

- insights
- simulator predictions
- weekly summaries
- score calculations

Clinically supported associations:

Tg-lowering habits:

- **Alcohol ↓↓↓**
- **Weight loss ↓↓**
- **Carb restriction ↓↓**
- **Daily steps / cardio ↓↓**
- **Reducing late-night eating ↓**
- **Omega-3 intake ↓**

LDL-lowering habits:

- **Saturated fat ↓↓↓**
- **Fiber ↑↑**
- **Plant sterols ↑**
- **Weight loss ↓**
- **Statins ↓↓↓↓↓ (powerful)**
- **Replacing butter with olive oil ↓**

Your app will gently reflect these in insights.

⭐ STEP 4 — SCREEN LIST (WIREFRAME OUTLINE)

These will look clean, modern, TotTrackr-style.

1. Onboarding

- Choose LDL / TG mode

- Basic demographics (age, sex)
- Initial labs (optional)
- Explanation of Cholesterol Score

2. Home Screen

- Big Cholesterol Score
- Daily Behavior Tracker (quick taps)
- "Today's Heart Score"
- Mini insight ("Steps help TG")

3. Labs Screen

- Add Lab + Camera button
- List of past labs
- Chart (line graph)
- Trend arrows (green/red)

4. Patterns Screen

- Behavior vs LDL chart
- Behavior vs TG chart
- Weekly summary card
- Recommendation engine

5. Food Guidance

- Green list (LDL-friendly)
- Green list (TG-friendly)
- Red list
- Better swaps
- Search bar ("Is shrimp okay?")

6. Settings

- Switch focus mode
- Reminders
- Export
- About
- Notifications

After this, you'll have a fully spec'd UI/UX to build.

⭐ STEP 5 — Build Plan (Flutter)

I'll deliver this in the next turn, but here's the summary:

Tech Stack:

- **Flutter** (fastest for you, cleanest UI)
- Firebase Lite or local storage for MVP
- OCR (via Google MLKit) v2
- Optional cloud function for insights

Architecture:

- Home → Behavior → Labs → Patterns → Food → Settings
- Simple state management (Provider or Riverpod)
- SQLite for local data logging

⭐ STEP 6 — Name + Branding (Later)

We'll pick between:

- LipidLog (clean, serious)
- Trigly (fun, modern)
- CholestCheck (straightforward)

- LipidLens (visualized health)

Or create variants.

⭐ STEP 7 — Monetization (since you're ALWAYS interested)

The cholesterol niche monetizes EXTREMELY well because users are:

- older
- health-conscious
- financially stable
- motivated
- willing to pay for outcomes

Recommended monetization strategy:

🟣 1. Freemium + Subscription + Lifetime Deal Hybrid

Best structure:

Free Tier:

- Manual lab entry
- Cholesterol Score
- Daily habit tracker
- Basic insights
- Food guidance

Pro Tier: $4.99/mo or $29/year

Unlocks:

- OCR lab import
- Weekly AI insights
- Advanced patterns
- What-If Simulator
- Apple Health sync
- Custom reminders
- Exporting reports for doctors
- Unlimited lab history

Lifetime Deal: $49–$69

Launch promo.

You know this works — it drives early revenue.

🟣 2. Add-on Upsells Later

- Meal scan (AI)
- Personalized diet plan
- Supplement tracking
- Medication interaction insights
- Dedicated triglyceride module

🟣 3. Long-term Monetization

Possible expansions:

- Android release
- Doctor portal (providers pay)
- Corporate wellness

⭐ Ready for Step 5?

Do you want me to:

👉 **Continue with the Build Plan (Step 5)**

or

👉 **Refine the MVP / Score / Habits first?**

Just tell me: **"Continue."**

The MVP is TIGHT let's look the build plan. Quick question: why do you suggest Flutter over React Native? Any particular reason?

Thought for ... >

Love it. Let's get into the build plan 🚀 🗓️

First I'll answer the Flutter vs React Native thing, then I'll lay out a concrete plan you can follow.

❓ Why I leaned toward Flutter over React Native

Given *you specifically*:

- You've **already shipped stuff in Flutter** (MySocialBattery), so you're past the "what is this" stage.
- You're a **vibe coder**, not trying to wrestle with Xcode / Gradle / native modules any more than necessary.
- This app is **very UI-heavy** (custom cards, charts, nice animations) but not "deeply native" (no Bluetooth, AR, etc).

So, practically:

Flutter pros for this project

1. **Pixel-perfect UI on both platforms**
   - Flutter owns the entire rendering pipeline; you're not fighting "why does this look different on Android vs iOS?"
   - Nice for charts, cards, gradients, custom score gauges.
2. **Widget-first mental model**
   - Everything is a widget (cards, lists, screens).
   - For a dashboard-style app, this maps really cleanly to how you think about the UI.
3. **Packages you'll actually need are mature**
   - Local storage (Hive / sqflite)
   - Charts (fl_chart, syncfusion_flutter_charts)
   - OCR (google_ml_kit)
   - State management (Riverpod / Provider)
     These are all well-trodden paths in Flutter land.
4. **Less "ecosystem fragmentation drama"**
   - With RN you can get stuck in:
     - "Which navigation library?"
     - "Which gesture handler?"
     - "Which chart lib works with the latest RN version?"
   - Flutter's ecosystem is smaller but more coherent, which is easier for a solo builder.

React Native pros (for fairness)

- You already know JS/TS from your other projects.
- If you ever want a **shared logic core** between web (Next.js) + mobile, RN + shared TS libs can help.
- If you really want to lean on web-style thinking (Flexbox, JSX), RN feels familiar.

My recommendation for *this* app

If the goal is: "**Get a clean, stable, pretty iOS + Android app in the store that I can maintain solo**" →
I'd still nudge you to **Flutter**.

If you said: "**I 100% want a web version next year that shares logic with the mobile app**" → I'd lean you towards **React Native + TypeScript**, and we'd design the logic layer shared.

You're not locked in by this decision forever, but it *will* affect how frustrating the build feels.

🛠️ Build Plan for the MVP (assuming Flutter)

I'll keep this concrete and step-wise so you can literally check things off.

Phase 0 — Setup (½–1 day)

- Install Flutter SDK, Android Studio or VS Code + Flutter extensions.

- Create project: `lipidlog` or `trigly`
- Add baseline dependencies in `pubspec.yaml`:
  - `flutter_riverpod` (state mgmt) or `provider`
  - `hive` + `hive_flutter` (or `sqflite`) for local storage
  - `fl_chart` (charts)
  - `google_ml_kit` (for later OCR, can be added later)
  - `intl` (date formatting)

Folder structure:

```
lib/ main.dart app.dart core/ models/ services/ theme/ features/
onboarding/ home/ labs/ habits/ patterns/ food/ settings/ widgets/
```

Phase 1 — Data Model + Storage (1–2 days)

Define core models (simple Dart classes):

- `UserProfile`
  - age, sex, focusMode (LDL / TG / Both)
- `LabResult`
  - id, date, ldl, hdl, triglycerides, total, nonHdl, fastingFlag
- `DailyLog`
  - date
  - alcoholLevel (0/1/2+)
  - stepsGoalHit (bool)
  - lateNightEating (bool)
  - highCarbDay (bool)
  - highSatFatDay (bool)
  - sleepCategory (short/normal/long)
  - medTaken (bool)
  - stressLevel (low/med/high)
- `ScoreSnapshot`
  - date, overallScore, labScore, habitScore, trendScore

Wire them up with Hive (or sqflite):

- Initialize on app start.
- Boxes/tables: `user_profile`, `labs`, `daily_logs`, `score_snapshots`.

Goal for this phase:

You can **save and read** labs + daily logs from local storage. No UI polish needed yet.

Phase 2 — Navigation + Screen Skeletons (1–2 days)

Use Flutter's `Navigator` or `go_router`.

Screens:

- `OnboardingScreen`
- `HomeScreen`
- `LabListScreen` + `AddLabScreen`
- `DailyLogScreen` (or bottom sheet)
- `PatternsScreen`
- `FoodScreen`
- `SettingsScreen`

At this point, just scaffold:

- App bar titles
- Basic layout
- Bottom navigation bar:
  - Home
  - Labs
  - Patterns
  - Food
  - Settings

No real logic yet, just wiring navigation.

Phase 3 — Lab Entry + Cholesterol Score Engine (core logic) (3–5 days)

1. **AddLabScreen**
   - Form fields for LDL, HDL, TG, Total, Non-HDL, date, fasting toggle.
   - On save: write `LabResult` to storage.
2. **Score Engine service**

   Create `core/services/score_service.dart`:
   - Function `calculateLabScore(List<LabResult> labs)` → returns 0–50.
   - Function `calculateHabitScore(List<DailyLog>)` → returns 0–30.
   - Function `calculateTrendScore(List<LabResult>)` → returns 0–20.
   - Function `calculateOverallScore(...)` → 0–100.

   For now you can hard-code the ranges and weights we discussed; later, refine.
3. **HomeScreen**
   - Reads latest `ScoreSnapshot` from storage.
   - Shows big circle or gauge with the score.
   - Shows last lab values underneath.
4. Whenever labs or daily logs are updated:
   - Recalculate the latest score.
   - Save a `ScoreSnapshot` for that date.

Goal:

You can **enter labs and see a meaningful Cholesterol Score** update.

Phase 4 — Daily Habit Logging (2–4 days)

1. **DailyLog UI**
   - On Home, a "Today's Habits" card with buttons:
     - Alcohol: [0] [1] [2+]
     - Steps: [Not Hit] [Hit]
     - Sleep: [<6] [6–8] [>8]
     - etc.
   - Use toggles / chips for fast, one-handed logging.
2. On save:
   - Create or update `DailyLog` for today.
   - Re-run score calculation.
   - Show small success message: "Great! Logging your day helps refine your Cholesterol Score."
3. **Weekly Habit Summary (simple)**
   - In Patterns screen, show:
     - Alcohol days this week
     - Steps goal hit %
     - Med adherence %
   - This will later feed into AI/text insights.

Goal:

You can **log habits daily in under 10 seconds** and see them reflected in the score.

Phase 5 — Trends & Patterns Screen (2–4 days)

1. **Charts**
   - Use `fl_chart` to draw:
     - Line chart: LDL over time.
     - Line chart: TG over time.
     - Line chart: Cholesterol Score over time.
2. **Simple correlation logic (MVP)**
   - For each week, compute:
     - average alcohol level
     - steps hit percentage
     - average TG
   - Show templated insights like:
     - "Weeks with fewer than 2 alcohol days had lower average TG."
     - "Your Score improved during weeks with ≥4 days of steps goal."

   For MVP you can do simple threshold checks; AI/NLP can come later.

Goal:

Patterns screen tells a story, even if the logic is basic at first.

Phase 6 — Food Guidance (1–3 days)

Keep it dead simple; don't try to build MyFitnessPal.

1. Hard-code or store a small JSON of food groups:
   - LDL-friendly
   - TG-friendly
   - Avoid / limit
   - Better swaps (butter → olive oil, etc.)
2. `FoodScreen`
   - Tabs: LDL / TG / Swaps.
   - Simple cards list.
   - Search bar filtering the local list.

Later you can expand this or integrate with a proper database, but MVP only needs a **reference, not a tracker**.

Phase 7 — Settings, Polish, Monetization Hooks (3–5 days)

Settings:

- Switch focus mode (LDL / TG / Both).
- Reminder toggles:
  - Daily habit log reminder.
  - Lab check reminder (3–6 months).
- Export: share CSV of labs + scores via email/files.

Monetization preparation:

- Wire in basic paywall screen (even if you don't enable purchases yet):
  - Free vs Pro features list.
  - "Upgrade to Pro" button.
- Plan for:
  - Pro-only: OCR import, What-If simulator, advanced insights, Apple Health sync.
- Decide prices:
  - e.g., `$4.99/mo`, `$29/yr`, `$59 lifetime`.

You don't have to implement IAP on day one, but **design the app with a clear Free vs Pro separation** so it's easy later.

📃 Tiny To-Do Summary (for you)

1. ✅ Choose Flutter vs React Native (I'd say Flutter for you).
2. Set up project + basic dependencies.
3. Implement models + local storage.
4. Scaffold navigation + screens.
5. Build lab entry + score engine.
6. Add daily habit logging.
7. Implement trends/patterns with basic charts + insights.
8. Add simple food guidance.
9. Wire settings + future paywall structure.
10. THEN: refine UI, branding, and IAP.

If you want, next we can:

- Flesh out the **exact data ranges/weights** for the Score engine so it feels legit, *or*
- Start on **wireframe-level UI** for each screen (with text labels and rough component layout), *or*
- Draft your **Free vs Pro feature breakdown + paywall copy** since you're monetization-brained.

Tell me which one you want to tackle next and we'll keep stacking bricks.

Minor enhancement suggestions from gemini

Enhance #3 - Cholesterol Score (The emotional hook): The 0-100 score is excellent. For the MVP, also prominently feature a clear, personalized Goal Score.

Example: If their doctor is aiming for an LDL under 100, the app should calculate what score (e.g., 85) reflects that goal being met.

Value: It shifts the score from "how good am I doing?" to "how close am I to my goal?"—a much stronger motivator.

Tier 2 - Lightweight Enhancements (Suggestion)
Move #3 (Medication list + adherence tracker) to Tier 1.

Reasoning: Medication adherence (statins, fish oil, etc.) is one of the most powerful variables in your Behavior Score calculation, and it directly influences the patient/doctor conversation. Making this a Tier 1 feature strengthens the Behavior Score (B) and the Trends Score (C) from day one.

These are **really good enhancements** — and they align perfectly with both *behavioral design* and *clinical reality*. Let's evaluate them and then weave them into the MVP cleanly.

✅ Enhancement #1 — Add a "Goal Score" to the Cholesterol Score Section

🔥 Why this is brilliant

Your score (0–100) is a **motivation engine**, but adding a *Goal Score* turns it into a **progress engine**.

Behavioral science:
People are more motivated by *closing a gap* than by hitting an abstract "good" number.

Clinically:
Every patient has a **target LDL** or **target TG**, but no app translates that into something intuitive or emotionally resonant.

🚀 How to implement (MVP-safe)

1. Let the user set a target LDL / TG (optional)

During onboarding, ask:

"Has your doctor given you a target LDL or TG level?"

- LDL target: 70 / 100 / 130 / custom
- TG target: 150 / 200 / custom
- "I don't know" → app suggests the standard guideline

2. Convert this into a Goal Score

Your Cholesterol Score algorithm already knows:

- LabScore = up to 50 points
- HabitScore = up to 30 points
- TrendScore = up to 20 points

We create a **Goal Score** = the score the user will reach *when their lab target is met* (with average habits and trend-neutral).

Example:

- Current Score: **61**
- Goal Score: **84**
  (This represents achieving LDL < 100 or TG < 150)

3. UI Display (emotional hook)

Show on Home:

Your Score: 61
Goal: **84**
You're **23 points away** from your heart-health target.

Or a progress bar:

```
▓▓▓
```

This is massively motivating.

🌟 Verdict:

Absolutely Tier 1.
Simple to implement, huge psychological payoff.

✅ Enhancement #2 — Move Medication Adherence (currently Tier 2) into Tier 1

🔥 Why this is correct

For LDL:

- Statins reduce LDL by **30–50%**
- Ezetimibe reduces LDL by **20–25%**
- PCSK9 inhibitors reduce LDL by **50–60%**

For triglycerides:

- Prescription fish oil (EPA) reduces TG by **25–45%**
- Fibrates reduce TG by **35–50%**

Translation:

Medication adherence is the **single strongest modifiable input** in your entire scoring model.

So if the app ignored meds until Tier 2, your MVP would be missing a key predictor of lipid change.

💊 Why include it Day One:

1. **You'll serve older users too** — many are on statins.
2. **Medication → behavior → lab changes** is central to cholesterol health.
3. The app becomes more useful during doctor visits.
4. It lets the AI/pattern engine make accurate insights:
   - "Weeks when you took your statin daily saw a 10% LDL drop trend."
5. It improves the TrendScore dramatically.

🚀 What Tier 1 medication tracking should look like (lightweight)

No need for a huge medication library.

Just:

- Statin (yes/no, name optional)
- Fish oil / EPA supplement (yes/no)
- Fibrate (optional)
- "Medication taken today?" (checkbox on habit card)
- Reminder toggle (optional MVP)

That's it.

🌟 Verdict:

Move medication adherence to Tier 1.
Make it part of the daily habit logging.
Huge improvement.

🔧 Updated MVP Feature List (with Gemini's enhancements merged)

Tier 1 (Core MVP)

1. Onboarding with LDL/TG mode
2. Manual lab entry + date + fasting
3. *Cholesterol Score (0–100)*
4. **New: Goal Score** based on LDL/TG target
5. Daily habit taps
6. **Medication adherence (moved up)**
7. Trend analysis
8. Weekly summary (templated)
9. Simple food guidance
10. Settings (reminders, focus mode)

Tier 2 (Post-launch Enhancements)

- Full medication library

- OCR for lab reports
- What-If Simulator
- Apple Health integration
- AI-generated insights
- Meal scan logging
- Correlation engine v2
- Export-to-doctor PDF report
- Weight and waist tracking
- Sleep integration

🔥 Summary: These enhancements tighten your MVP and sharpen your differentiation.

With:

- **Cholesterol Score**
- **Goal Score**
- **Medication adherence**
- **Daily habits**
- **Trends**
- **One gorgeous UI**

You would *immediately* be the best lipid-health app on the App Store.

Next Steps

Do you want me to:

👉 **Incorporate these into the build plan**
👉 **Refine the Score algorithm in detail (with new Goal Score logic)**
👉 **Start designing screen wireframes (Figma-style)**
👉 **Move to monetization modeling**

Just tell me what you want next.

> Refine the algorithm then incorporate into the build plan then wireframe

Thought for ...

Love this flow. Let's tighten the brain of the app first, then wire it into the build plan, then sketch the screens.

1️⃣ Refined Cholesterol Score Algorithm (with Goal Score + Meds)

> Not medical advice, just a structured way to turn your labs + habits into a motivational score.

We'll keep the **100-point score**, broken into:

- **LabScore** – 0–50
- **BehaviorScore** – 0–30
- **TrendScore** – 0–20

Then add:

- **GoalScore** – "where your Score will be when you hit your lab target"

A. LabScore (0–50)

We normalize 5 lab metrics to 0–10 each, then weight them.

1. Normalization functions (v1)

You can put these as simple `if/else` helpers in `score_service.dart`.

LDLScore (0–10)

```
LDL (mg/dL) Points ≤ 70 10 71–99 9 100–129 7 130–159 5 160–189 3 ≥ 190 0
```

TGScore (0–10)

```
Triglycerides Points < 100 10 100–149 9 150–199 7 200–499 4 ≥ 500 0
```

HDLScore (0–10) — higher is better

(same table for now; you can later adjust for sex)

```
HDL (mg/dL) Points ≥ 60 10 50–59 8 40–49 6 < 40 2
```

RatioScore (Total / HDL)

```
Ratio Points ≤ 3.5 10 3.6–4.5 8 4.6–6.0 5 > 6.0 2
```

NonHDLScore (0–10)

```
Non-HDL Points < 100 10 100–129 8 130–159 6 160–189 4 ≥ 190 1
```

2. Lab weighting

Weights:

- LDL: **35%**
- TG: **25%**
- HDL: **15%**
- Ratio: **15%**
- Non-HDL: **10%**

In **LDL mode**, LDL & Non-HDL matter more; in **TG mode**, TG gets more weight. But for v1, you can use this constant set and tweak later.

```
weighted10 = 0.35 * ldlScore10 + 0.25 * tgScore10 + 0.15 * hdlScore10 +
0.15 * ratioScore10 + 0.10 * nonHdlScore10; LabScore = clamp(weighted10 *
5, 0, 50); // 0–50
```

B. BehaviorScore (0–30) with Med Adherence

We treat each day as a 0–10 "behavior points" day, then average over the last 14–30 days and scale to 0–30.

1. Daily behavior components (raw points)

For each day (pulled from `DailyLog`):

- **Steps goal hit**
  - yes → +2
  - no → 0
- **Alcohol**
  - 0 drinks → +2
  - 1 drink → +1
  - 2+ drinks → –2 (heavier hit for TG mode)
- **Sleep**
  - 6–8h → +1
  - 8h → +0.5
  - <6h → –1
- **Late-night eating**
  - yes → –1
  - no → 0
- **High-carb day** (self-reported)
  - yes → –1 (weight more in TG mode later)
  - no → 0
- **High saturated fat**
  - yes → –2 (especially important in LDL mode)
  - no → 0
- **Medication taken** (statin / TG meds / fish oil etc.)
  - yes → +2
  - no → –1
- **Stress**
  - high → –1
  - med → 0
  - low → +0.5

So a day might range roughly from **–4 to +10**.

2. Normalize to 0–10

```
rawDaily = sum(componentPoints); // from -4 to +10-ish clamped =
clamp(rawDaily, -4, 10); normalizedDaily10 = ((clamped + 4) / 14) * 10; //
maps -4..10 → 0..10
```

3. Aggregate → BehaviorScore 0–30

Take the average of the last N days (say 14 or 30):

```
avgDaily10 = average(normalizedDaily10 over last N days); BehaviorScore =
clamp(avgDaily10 * 3, 0, 30);
```

C. TrendScore (0–20)

We look at **change over time** in the *focus metric*:

- LDL in LDL mode
- TG in TG mode
- Maybe Non-HDL or ratio in "Both" mode (v2)

Use the oldest vs latest of the last 3 labs (or last 6–12 months, whichever is shorter).

```
if less than 2 labs: TrendScore = 10 // neutral, we don't know yet else:
delta = latest - oldest pctChange = delta / oldest * 100
```

Map pctChange to points:

```
Change in LDL/TG TrendScore ≤ −20% 20 −20% to −10% 16 −10% to −5% 12 −5%
to +5% 10 +5% to +15% 5 > +15% 0
```

D. Overall Score (0–100)

```
OverallScore = LabScore + BehaviorScore + TrendScore; OverallScore =
clamp(OverallScore, 0, 100);
```

That's what you display as the big ring/gauge.

E. GoalScore (personalized target)

GoalScore answers: "**What score corresponds to hitting my doctor's target?**"

1. User inputs / we suggest targets

During onboarding or in Settings:

- LDL target: 70 / 100 / 130 / custom
- TG target: 150 / 200 / custom

Depending on mode:

- LDL mode → use LDL target
- TG mode → use TG target
- Both → you can use the stricter of the two or calculate both and average.

2. Compute LabScore at target

We reuse our lab scoring but **substitute the target value** for the relevant metric, keep the others at latest values:

```
latestLabs = most recent LabResult if mode == LDL: targetLabs = latestLabs
but with LDL = LDL_target else if mode == TG: targetLabs = latestLabs but
with TG = TG_target // later: handle 'Both' as combo LabScore_goal =
LabScore(targetLabs) // using same function as overall
```

3. Assume neutral behavior & trend at goal

To keep it simple for MVP:

- BehaviorScore at goal → **15** (half of max 30)
- TrendScore at goal → **10** (neutral)

You can tweak these later, but this works.

```
GoalScore = clamp(LabScore_goal + 15 + 10, 0, 100);
```

4. Distance-to-go & UI

```
deltaToGoal = GoalScore - OverallScore; // negative means you've surpassed
it
```

In the UI:

- "Your Score: **61**"

- "Goal: **84**"
- "You're **23 points away** from your target."

And in weekly summary:

- "You moved **4 points closer** to your Goal Score this week."

2️⃣ Incorporate the Algorithm into the Build Plan

Here's how this plugs into the build steps you already have.

Phase 1 — Models (unchanged conceptually)

Add fields to support targets:

- `UserProfile`
  - `ldlTarget` (nullable)
  - `tgTarget` (nullable)
  - `focusMode` enum (LDL, TG, BOTH)
- `ScoreSnapshot`
  - `overallScore`
  - `labScore`
  - `behaviorScore`
  - `trendScore`
  - `goalScore` ✅ new
  - `deltaToGoal` ✅ new

Phase 3 — Score Engine implementation

In `core/services/score_service.dart` (or similar):

- `double computeLDLScore10(double ldl);`
- `double computeTGScore10(double tg);`
- `double computeHDLScore10(double hdl);`
- `double computeRatioScore10(double total, double hdl);`
- `double computeNonHDLScore10(double nonHdl);`
- `double computeLabScore(List<LabResult> labs, FocusMode mode);`
- `double computeDailyBehavior10(DailyLog log, FocusMode mode);`
- `double computeBehaviorScore(List<DailyLog> logs, FocusMode mode);`
- `double computeTrendScore(List<LabResult> labs, FocusMode mode);`
- `double computeGoalScore(UserProfile profile, LabResult latest, FocusMode mode);`
- `ScoreSnapshot computeScores(UserProfile profile, List<LabResult> labs, List<DailyLog> logs);`

The `computeScores` function becomes your **single entry point**:

```
ScoreSnapshot computeScores(profile, labs, logs): labScore = computeLabScore(labs, profile.focusMode) behaviorScore = computeBehaviorScore(logs, profile.focusMode) trendScore = computeTrendScore(labs, profile.focusMode) overall = clamp(labScore + behaviorScore + trendScore, 0, 100) latest = most recent lab (if any) if latest and profile has target: goal = computeGoalScore(profile, latest, profile.focusMode) else: goal = null (UI hides it or shows "Set goal") delta = (goal != null) ? goal - overall : null return ScoreSnapshot( date: today, overallScore: overall, labScore: labScore, behaviorScore: behaviorScore, trendScore: trendScore, goalScore: goal, deltaToGoal: delta, )
```

Where to call it

- After any **lab entry** is saved.
- After any **daily log** is updated.
- After user **changes target** or **focus mode**.

Then HomeScreen just listens to the latest `ScoreSnapshot`.

3️⃣ Wireframes: How the Screens Use the Score

Let's keep these Figma-style but text-based so you can picture them.

🏠 Home Screen – "Today"

Top section

- Big circle / gauge:

- Center:
  - `Score: 61`
  - Small text: `Goal: 84`
  - Under: `23 points to your target`
- Subtext:
  - `LDL: 142 • TG: 260 • HDL: 42` (last lab)

Middle section – "Today's Habits"

Card with quick-tap chips:

- **Medication**
  - [Took my meds ✅] [Missed ❌]
- **Steps**
  - [Hit goal] [Not yet]
- **Alcohol**
  - [0] [1] [2+]
- **Sleep**
  - [<6h] [6–8h] [>8h]
- **Food**
  - [High-carb today] [High sat-fat today]
- **Other**
  - [Late-night eating] toggle
  - [Stress: Low / Med / High]

Save button or auto-save on tap.

Bottom section

- "This week so far"
  - Horizontal progress showing:
    - Days meds taken
    - Days steps goal hit
    - Alcohol-free days

🧪 Labs Screen

Header

- Title: `Labs`
- Button: `+ Add Lab`

Content

- Latest lab summary card:
  - `Date`
  - LDL / HDL / TG / Total / Non-HDL
  - Small badge: `Fasting / Non-fasting`
- Chart:
  - Tabs: [LDL] [TG] [Score]
  - Line chart for selected metric over time
- List:
  - Past labs as simple rows by date
    tapping opens details / edit.

📊 Patterns Screen

Top

- Subtitle: `Patterns & Trends`

First card: Trend

- "Your LDL trend:"
  - Arrow + text: `-12% over last 6 months`
  - `TrendScore: 16/20`

Second card: Behavior vs labs

- Simple text insights generated from thresholds, e.g.:
  - "Weeks with **0–1 alcohol days** had **18% lower average TG**."
  - "On weeks you hit your **steps goal ≥4 days**, your Score improved by **+6 points**."

Chart section

- Small chart showing:
  - Bars: Steps-goal days per week

- Line: Cholesterol Score per week

🥦 Food Screen (MVP Utility, not a full tracker)

Tabs at top:
`[LDL-friendly] [TG-friendly] [Swaps]`

List of cards:

- `Extra virgin olive oil` – "Great butter replacement for LDL."
- `Fatty fish` – "Helps triglycerides with omega-3s."
- `Refined carbs` – "Limit to support TG control."

Search bar at top:
"Search foods…"

⚙️ Settings Screen

Sections:

Profile

- Age, sex
- Focus mode: LDL / TG / Both

Targets

- LDL target (picker / custom)
- TG target (picker / custom)
- Button: "Explain targets" (simple explainer, non-medical)

Reminders

- Daily habit log reminder (time picker)
- Med reminder
- Lab check reminder (every 3 / 6 / 12 months)

Data & Pro

- "Export my data (CSV / JSON)"
- "Upgrade to Pro" (later: paywall)